

TrES-5b

R-1306

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_141026 · created 2026-07-31T17:59:15+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456956.6251 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.108 +/- 0.044
Transit depth (Rp/Rs)^2	1.16 +/- 0.94 [%]
Orbital Inclination (inc)	89.45 +/- 2.96
Airmass coefficient 1 (a1)	1.004 +/- 0.023
Airmass coefficient 2 (a2)	0.0 +/- 0.052
Scatter in the residuals of the lightcurve fit is	2.3 %
Best Comparison Star	#1 - [582, 154]
Optimal Aperture	3.03
Optimal Annulus	6.27
Transit Duration (day)	0.0799 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.108 ± 0.044 vs 0.142 (deviation 0.73σ)
Duration fit vs published	0.0799 d vs 0.0758 d (deviation 0.65σ)
Mid-time O–C	-2.9 min
Depth SNR	2.3
Residual scatter	2.3 %

Light curve

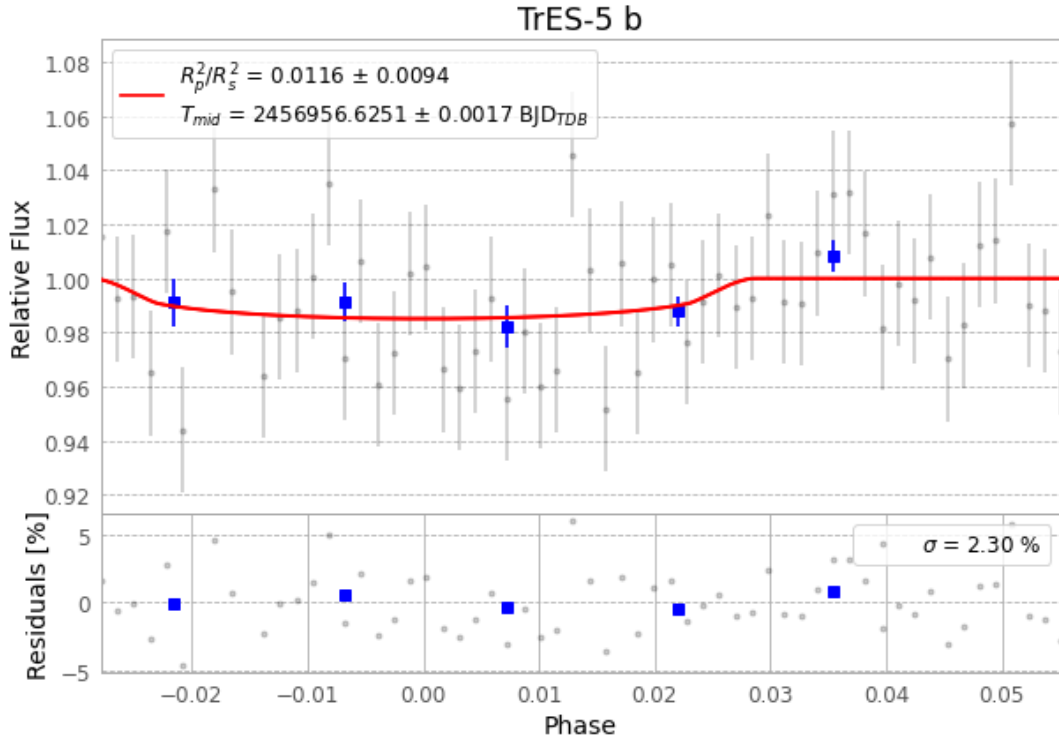


Figure 1 — FinalLightCurve_TrES-5 b_2014-10-25.png (SHA-256 3df6150d591bef81...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [383, 354], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bad0a9a567e59e08...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

60 FITS frames (39 MB), TRES-5141026015712.FITS → TRES-5141026045412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-141026.log (565f65f1b86f...), FinalLightCurve_TrES-5 b_2014-10-25.png (3df6150d591b...), FinalLightCurve_TrES-5 b_2014-10-25.csv (d87681426265...)

Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 17:59Z.